# Chethan Reddy

Email: achethanreddy1921@gmail.com
LinkedIn — GitHub — Portfolio — Twitter — LeetCode — Coursera
Mobile: +91-9030496717
Forward Deployed Engineer (FDE-2) — Agentic AI — Applied AI Systems

## Technical Experience

- **Staple AI — Forward Deployed Engineer (FDE-2)** — Singapore, Hybrid
  *Current role: May 2026 – Present* — *At Staple: May 2024 – Present*
  - **Solution architecture & delivery:** Own technical scoping and delivery for **20+ enterprise customers**; turn discovery findings into implementation plans with product engineering, customer success and QA.
  - **Adoption agents & customer intelligence:** Build AI agents and dashboards around customer telemetry to surface usage patterns and upsell opportunities; support customer adoption and usage-growth discussions.
  - **Customer enablement:** Trained users on extraction/data quality; wrote API guides and implementation handovers.
  - **Production reliability:** Investigated ingestion and extraction failures using logs, document comparisons and root-cause analysis; led transactional repairs and monitored recovery with dry-run tooling and explicit verification checks.

  ***Product engineering contributions across SDE/FDE roles***
  *Earlier: AI Software Engineer — SDE-2 from May 2025; SDE-1 from May 2024.*
  - **AI feature delivery:** As a core engineer, shipped complex-table extraction across **3 services and 3 regions**; built Python APIs, model/template schemas and per-page mapping of extracted cells to OCR boxes.
  - **Distributed systems:** Co-designed and implemented async document processing with RabbitMQ, retries, deduplication and dead-letter handling; tested scan-message compatibility against **12 reference fixtures** in staging.
  - **LLM evaluation:** Built a Next.js/FastAPI platform across **8 providers and 25 models**, from requirements through staging handoff; compared ground-truth F1/similarity, repeat-run consistency and token cost, with JSON validation and model routing.
  - **QA automation:** Built regression tests for **70 API endpoints** with response contracts, negative cases and isolated-data cleanup; moved region-aware scheduling to GitHub Actions using Postman CLI.
  - **Shared infrastructure:** Created a Python library for telemetry, database sessions and S3 access with **165 tests**; separated read/write traffic across **6+ services** with replicas, connection-pool tuning and distributed tracing.
  - **Customer migrations:** Authored migration playbooks with baseline snapshots, extraction/export comparisons and rollback procedures; built dry-run tooling, idempotency checks and post-write verification for controlled releases.

- **Dotnitron Technologies — AI Software Engineer** — USA, Remote
  *Alvarez & Marsal engagement: Diligence GPT* — *Dec 2023 – May 2024*
  - **RAG architecture:** Designed retrieval-augmented generation workflows for Diligence GPT with cross-functional teams, connecting document retrieval and AI-generated analysis for enterprise due-diligence use cases.
  - **Backend leadership:** Led Python backend development with **Django, FastAPI and ArangoDB**, integrating AI tools and libraries into the application's services and database-backed retrieval workflows.
  - **AI integration and quality:** Applied LangChain and OpenAI to backend features; reviewed code, debugged integrations and tested AI workflows. The Alvarez & Marsal client engagement continued through June 2024.

- **Bajaj Finserv Health — Data Science Engineer Intern** — Pune, India
  *InSightRx Data Team* — *May – Aug 2023*
  - **Recommendation systems:** Led development of user-friendly lab-test recommendations for the InSightRx data team, improving operational efficiency by **30%** in project measurements.
  - **Healthcare image analytics:** Applied **LLMs, OCR, NER and YOLO** to healthcare documents and images, combining text recognition, named-entity extraction and object detection to support downstream data analysis.
  - **Search optimization:** Refined Elasticsearch and SQL queries with fuzzy matching to improve access to healthcare information, achieving **80% faster retrieval** in project measurements.

- **TIFAC / VIT — Backend Development & Data Intern** — Vellore, India
  *Backend Lead, Physio Plus* — *Dec 2022 – Jun 2023*
  - **Backend ownership:** Led the complete Physio Plus backend using **FastAPI and MongoDB**; designed patient-record and clinic workflows for MicroClinics and supported analysis of patient data.
  - **Scale and infrastructure:** Used AWS and a distributed architecture to support **100+ daily patient appointments**; also researched algorithms for autonomous drones in inventory management.

## Education

- **Vellore Institute of Technology** — Vellore, India
  *Bachelor of Engineering, Computer and Electrical Engineering*

## Earlier Experience and Mentoring

- **Smart Diet Planner — Machine Learning Researcher** Remote
  *Research and development internship* *Aug 2022 – Apr 2023*
  - **AI features:** Built search engines and automated PDF analysis with text and nutrition extraction for diet planning.
  - **Nutrition workflows:** Built macro-consumption calculations and nutrition analysis for personalized diet planning.
  - **Conversational AI:** Developed a customized **PaLM-based diet chatbot** for an application serving **850k+ users**, generating diet recommendations tailored to medical conditions and complications.
- **Grroom — Machine Learning Intern / Team Lead** India
  *Promoted to Team Lead in January 2022* *Nov 2021 – Feb 2022*
  - Trained YOLO for **100k+ iterations** for fashion-image detection and tagging, integrated models with the backend and led **10+ interns** developing ML algorithms and Python annotation scripts.
- **Tekie / Cybeorg / CipherSchools & other platforms** Remote
  *Python, ML and Software Engineering Mentor* *2021 – 2023*
  - **Teaching and code review:** Mentored **60+ students** through **500+ hours** at Tekie in Python/C++ and problem-solving; debugged coding submissions. At CipherSchools, reviewed Python projects and recommended improvements.
  - **Scientific programming:** Built Cybeorg curricula and delivered **300+ hours** in programming mathematics, ML, data science and astronomy; applied NumPy, SciPy and pandas in research-support projects and practical lessons.
  - **Software and ML mentoring:** Covered ML, databases, system design and algorithms at edvi; built ML curricula at Modo Edulabs and taught model development and Python/AI at RoboGems (experience) and Camp K12.

## Selected Engineering Projects

- **eduAId / KaRmA / hAIr:** Built eduAId class analytics, note digitization and LangChain agents; KaRmA's Mistral/Llama SQL reports; and parsing, matching and interview-question features within the hAIr team.
- **Supplier Analytics / Insurance Search:** Analyzed **7m+ supplier records** using rating, cost and delivery data; built FastAPI search, autocomplete and recommendations with linear SVMs, Levenshtein distance and graph matching.
- **WeDio / FFCS Planner / PhotoBook:** Built **five-class CCTV disruption detection**, graph-based course timetables and image-to-PDF APIs with upload, resize and layout operations using FastAPI, Pillow and FPDF.

## Hackathon Achievements

- **Winner, 1st Prize** — Bitget U-30 Hackathon (2023).
- **Winner, 1st Prize** — IEEE Women in Engineering VIT WeHack 3.0, powered by Honeywell (2022).
- **1st Runner-up** — Axis Bankathon LLM 1.0 (2023).
- **1st Runner-up, 2nd Prize** — HackRx 3.0, Bajaj Finserv Health (2022).
- **Grand Finalist, Top 4 of 3,000+ teams** — GLC–HP Hackathon (2024).
- **Top 5 finalists** — Canara Bank DACOE-Thon (2024).
- **Grand Finalist, Top 7** — SirionLabs HackFest (2022); 2,000+ entries.
- **Top 7** — Walmart Sparkathon (2023).
- **Phase 2, Top 50 teams** — Samsung Solve for Tomorrow, IIT Delhi (2022); 18,000+ registrations.
- **Finalist** — Siemens Healthineers SHIFT Hackathon (2022).
- **Top 10 of 150+ teams** — JSSATEB Hackwell 3.0, powered by Honeywell.
- **Top 5** — Amadeus H4S; travel advisory using 3D graphs.
- **Top 10** — Make-a-thon; AI Sight for eye-condition screening.
- **Top 10** — AI Arena 1.0; data analytics with prompt engineering.
- **Top 100** — SmartIdeathon (2023); multimodal transportation algorithms.

## Technical Skills

**Agentic AI & LLMOps:** AI agents, RAG, prompt engineering, LLM/VLM integration, structured outputs, evaluation/routing and observability; LangChain, Milvus retrieval evaluation; Hermes Agent / OpenClaw setup and configuration.
**Architecture & leadership:** Solution architecture, customer discovery/scoping, cross-functional technical leadership, mentoring, code review and enterprise API integration.
**AI & model training:** TensorFlow/Keras, NumPy, SciPy, pandas; supervised learning, neural networks, model training and evaluation; YOLO object detection, computer vision, OCR, named-entity recognition, SVMs and regression.
**Software & platforms:** Python, SQL, TypeScript/JavaScript, Go, C++; FastAPI, Django, Flask, React/Next.js; PostgreSQL, MongoDB, ArangoDB, Elasticsearch; RabbitMQ/Temporal; AWS, Kubernetes, OpenTelemetry, CI/CD.
**Coursework:** TensorFlow Hub NLP; Computer Vision Basics; Getting Started with AWS Machine Learning; full record.
**Spoken languages:** English (fluent), Telugu and Hindi.